

Natalia Rabinovich

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EDUCATION

Rutgers University Honors College

New Brunswick, NJ

Bachelor of Science in Aerospace Engineering, Dean's List

May 2028

- **Relevant Coursework:** Fluid Mechanics, Thermodynamics, Mechatronics, CAD, Mech Mats, Calc I – V
- **Technical Training:** NASA ARSET Certificate, UNP Satellite Fabrication Course

RELEVANT EXPERIENCE

Lead Systems Engineer

Sep 2026 – Present

NASA L'SPACE Mission Concept Academy

Online

- Lead a 9-person multidisciplinary engineering team developing a robotic payload for lunar permanently shadowed regions, coordinating mechanical, electrical, and thermal design to reduce uncertainty for future human exploration.
- Develop ConOps and mission architecture while integrating subsystem designs into a single lunar surface payload.
- Completing Siemens NX Design Associate Certification through the Siemens Xcelerator Academy.

Thermal Engineering Project Lead

May 2026 - Present

SPICESat CubeSat, Rutgers

New Brunswick, NJ

- Lead a 4-member team developing the thermal-control strategy for a CubeSat fluid payload, translating 9 mission requirements into thermal limits, orbital analysis cases, and hardware design constraints for CDR readiness.
- Creating a 500+ node satellite model by simplifying spacecraft SolidWorks CAD and using Thermal Desktop to create surface contactors, assign material properties, and add surface heat loads and boundary nodes.
- Run 10+ hot/cold case orbital simulations in Thermal Desktop and use MATLAB to process temperature histories, predicting payload temperatures from -10C to 40C, and improving fluid temperature distribution gradient by 20%.
- Developing a Python data-acquisition script for physical conductance testing to log steady-state tank temperatures.

Program Manager

Sep 2024 – Present

SPICESat CubeSat, Rutgers

New Brunswick, NJ

- Lead an 8-subteam, 40-member CubeSat program, coordinating mechanical, thermal, electrical, and software teams through spacecraft design, hardware integration, engineering requirements, and industry-level technical reviews.
- Lead 15+ multidisciplinary design reviews per semester to identify subsystem interface conflicts, assign action owners, and resolve technical risks before hardware testing, submission deadlines, and major program reviews.

Undergraduate Research Assistant

Sep 2026 - Present

Space Propulsion with Advanced Chemistry and Energetics (SPACE) Lab, Rutgers

New Brunswick, NJ

- Conducting research in air breathing electric propulsion, working with VLEO wind tunnel, conducting simulations.

Mechanical Lead

Sep 2022 – May 2024

Questionable Engineering Team, FIRST Robotics Competition

Jersey City, NJ

- Led mechanical design in Onshape/AutoCAD, achieving reliable arm motion and object manipulation in matches.
- Supported on-the-spot design changes during the World Championship, awarded Rookie All Star and Highest Seed.

PROJECTS

Autonomous Docking Robot

Spring 2026

- Designed, built, and tested an autonomous docking robot using SolidWorks, 3D-printed hardware, an HC-SR04 sensor, DC motors, LEDs, and a servo-actuated clasp, using sensor to control speed and trigger docking sequence.

Mechanical Claw Machine

Spring 2026

- Designed a complete claw-machine assembly in SolidWorks, including the casing, gantry, claw, access door, shafts, guides, gears, and rack-and-pinion components needed for horizontal travel, vertical extension, and claw actuation.

SKILLS

Mechanical Design: SolidWorks, Onshape, AutoCAD, Fusion, FEA, CAD, 3D printing, mechanical prototyping

Hardware: Hand tools, soldering, wiring, staking, potting, conformal coating, cleanroom practices, hardware integration

Engineering: Trade studies, mechanical integration, technical documentation, PDR/CDR prep, root cause analysis

Programming: MATLAB, Python, C++, HTML, CSS, JavaScript

Software: Thermal Desktop, Microsoft Excel, PowerPoint, Word, Git, Linux

Languages: Fluent in English, Spanish, and Russian